

# Cryptology Assignment - 2

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1) Encrypt "CHASE YOUR DREAMS" using:

- Caesar cipher
- Shift Cipher with key 7
- ROT 13

|    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|
| A  | B  | C  | D  | E  | F  | G  | H  | I  | J  | K  | L  | M  |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 |
| N  | O  | P  | Q  | R  | S  | T  | U  | V  | W  | X  | Y  | Z  |
| 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

Caesar cipher:

Formula :  $CT = E(PT+3) \bmod 26$  (no. of alphabets)

|   |   |        |             |            |   |   |
|---|---|--------|-------------|------------|---|---|
| C | → | (2+3)  | mod 26 = 5  | mod 26 = 5 | = | F |
| H | → | (7+3)  | mod 26 = 10 | →          |   | K |
| A | → | (0+3)  | mod 26 = 3  | →          |   | D |
| S | → | (18+3) | mod 26 = 21 | →          |   | ✓ |
| E | → | (4+3)  | mod 26 = 7  | →          |   | H |
| Y | → | (24+3) | mod 26 = 1  | →          |   | B |
| O | → | (14+3) | mod 26 = 17 | →          |   | R |
| U | → | (20+3) | mod 26 = 23 | →          |   | X |
| R | → | (17+3) | mod 26 = 20 | →          |   | U |
| D | → | (3+3)  | mod 26 = 6  | →          |   | G |
| R | → | (17+3) | mod 26 = 20 | →          |   | U |
| E | → | (4+3)  | mod 26 = 7  | →          |   | H |
| A | → | (0+3)  | mod 26 = 3  | →          |   | D |
| M | → | (12+3) | mod 26 = 15 | →          |   | P |
| S | → | (18+3) | mod 26 = 21 | →          |   | V |

CipherText = FKDVH BRXU GVHDPV

## Shift Cipher (Key 7)

Formula :  $CT = E (PT + 7) \bmod 26$

|   |                                           |              |
|---|-------------------------------------------|--------------|
| C | $(P + 7) \bmod 26 = (2 + 7) \bmod 26 = 9$ | <del>J</del> |
| H | $(7 + 7) \bmod 26 = 14$                   | <del>D</del> |
| A | $(0 + 7) \bmod 26 = 7$                    | H            |
| S | $(18 + 7) \bmod 26 = 25$                  | Z            |
| E | $(4 + 7) \bmod 26 = 11$                   | L            |
| Y | $(24 + 7) \bmod 26 = 5$                   | F            |
| O | $(14 + 7) \bmod 26 = 21$                  | V            |
| U | $(20 + 7) \bmod 26 = 1$                   | B            |
| R | $(17 + 7) \bmod 26 = 24$                  | Y            |
| D | $(3 + 7) \bmod 26 = 10$                   | K            |
| R | $(17 + 7) \bmod 26 = 24$                  | Y            |
| E | $(4 + 7) \bmod 26 = 11$                   | L            |
| A | $(0 + 7) \bmod 26 = 7$                    | H            |
| M | $(12 + 7) \bmod 26 = 19$                  | T            |
| S | $(18 + 7) \bmod 26 = 25$                  | Z            |

Cipher Text = JOH ZL FVBY KYLHTZ

## ROT 13

Formula :  $CT = E(PT+13) \bmod 26$

|   |                                           |   |
|---|-------------------------------------------|---|
| C | $(PT+13) \bmod 26 = (2+13) \bmod 26 = 15$ | P |
| H | $(7+13) \bmod 26 = 20$                    | U |
| A | $(0+13) \bmod 26 = 13$                    | N |
| S | $(18+13) \bmod 26 = 5$                    | F |
| E | $(4+13) \bmod 26 = 17$                    | R |
| Y | $(24+13) \bmod 26 = 11$                   | L |
| O | $(14+13) \bmod 26 = 1$                    | B |
| U | $(20+13) \bmod 26 = 7$                    | H |
| R | $(17+13) \bmod 26 = 4$                    | E |
| D | $(3+13) \bmod 26 = 16$                    | Q |
| R | $(17+13) \bmod 26 = 4$                    | E |
| E | $(4+13) \bmod 26 = 17$                    | R |
| A | $(0+13) \bmod 26 = 13$                    | N |
| M | $(12+13) \bmod 26 = 25$                   | Z |
| S | $(18+13) \bmod 26 = 5$                    | F |

CipherText = PUNFR LBHE QERNZF

## 2) Decrypt:

a) WRWDOOB VHFUHW PHVVDJH

using Caesar cipher: Formula  $\Rightarrow PT = D(CT-3) \bmod 26$

$$W = (20-3) \bmod 26 = 19 = \underline{T}$$

$$R = 14 = \underline{O} \quad V = 18 = \underline{S} \quad P = 12 = \underline{M}$$

$$W = 19 = \underline{T} \quad H = 4 = \underline{E} \quad H = 4 = \underline{E}$$

$$D = 0 = \underline{A} \quad F = 2 = \underline{C} \quad V = 18 = \underline{S}$$

$$O = 11 = \underline{L} \quad U = 17 = \underline{R} \quad V = 18 = \underline{S}$$

$$O = 11 = \underline{L} \quad H = 4 = \underline{E} \quad D = 0 = \underline{A}$$

$$B = 24 = \underline{Y} \quad W = 19 = \underline{T} \quad J = 6 = \underline{G}$$

$$H = 4 = \underline{E}$$

Plain Text = TOTALLY SECRET MESSAGE



- b) MJQQT BTWQI using Shift Cipher (key 5)  
Formula:  $PT = (CT - 5) \bmod 26$

|                                             |                          |
|---------------------------------------------|--------------------------|
| $M = (12 - 5) \bmod 26 = 7 = \underline{H}$ | $B = 22 = \underline{W}$ |
| $J = 4 = \underline{E}$                     | $T = 14 = \underline{O}$ |
| $Q = 11 = \underline{L}$                    | $W = 17 = \underline{R}$ |
| $Q = 11 = \underline{L}$                    | $Q = 11 = \underline{L}$ |
| $T = 14 = \underline{O}$                    | $I = 3 = \underline{D}$  |

Plain Text = HELLO WORLD

- SJB DTWP YNRJ using shift Cipher (key 5)  
Formula:  $PT = (CT - 5) \bmod 26$

|                          |                          |                          |
|--------------------------|--------------------------|--------------------------|
| $S = 13 = \underline{N}$ | $D = 24 = \underline{Y}$ | $Y = 19 = \underline{I}$ |
| $J = 4 = \underline{E}$  | $T = 14 = \underline{O}$ | $N = 8 = \underline{I}$  |
| $B = 22 = \underline{W}$ | $W = 17 = \underline{R}$ | $R = 12 = \underline{M}$ |
|                          | $P = 10 = \underline{K}$ | $J = 4 = \underline{E}$  |

Plain Text = NEW YORK TIME

- c) FRAQ ZBER PBSSRR using ROT 13  
Formula:  $PT = (CT + 13) \bmod 26$

|                                              |                          |                          |
|----------------------------------------------|--------------------------|--------------------------|
| $F = (5 + 13) \bmod 26 = 18 = \underline{S}$ | $Z = 12 = \underline{M}$ | $P = 2 = \underline{C}$  |
| $R = 4 = \underline{E}$                      | $B = 14 = \underline{O}$ | $B = 14 = \underline{O}$ |
| $A = 13 = \underline{N}$                     | $E = 17 = \underline{R}$ | $S = 5 = \underline{F}$  |
| $Q = 3 = \underline{D}$                      | $R = 4 = \underline{E}$  | $S = 5 = \underline{F}$  |
|                                              |                          | $R = 4 = \underline{E}$  |
|                                              |                          | $R = 4 = \underline{E}$  |

Plain Text = SEND MORE COFFEE